



Research articles

Effect of yttrium substitution in Fe-doped ZnO nanoparticles: An EPR study

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ABSTRACT

In this work, we investigated the effect of Y^{3+} inclusion in ZnO nanoparticles doped with Fe^{3+} synthesized by the sol gel method. Transmission Electron Microscopy (TEM) images showed agglomerated nanostructures with different shapes and average particle size distributions ranging from 6 nm to 44 nm in diameter. Electron Paramagnetic Resonance (EPR) studies of the $Zn_{0.98-y}Y_{0.02}Fe_yO$ ($y = 0.00, 0.02, 0.03$ and 0.04) were carried out using X band (9.7 GHz) at room temperature. For undoped ZnO and doped with Y^{3+} , complex line shapes were observed as a result of oxygen vacancies ($V_O^{\cdot-}$). In addition, the Y^{3+} inclusion provokes an enhancement of the linewidth and the resonance field due to intrinsic defects in the hexagonal structure. This observation is compatible with a shortest relaxation time by reason of the particle size reduction and an extra antiferromagnetic local field as a result of the Y^{3+} presence. For co-doped ZnO samples, the EPR spectra confirms the presence of Fe^{3+} and Y^{3+} at the core Zn sites, altering the lattice strain of the ZnO structure. This result was monitored by the zero-field splitting (ZFS) parameters of the spin Hamiltonian, corroborating a reduction in the Fe - Fe interaction. The presence of Y ions was also confirmed in interstitial sites, resulting in a small g-shift from 4.2 to 4.0 associated with modifications of the rhombic component, E, of the Spin-Hamiltonian. This study contributes to the understanding of structural defects caused by the Y and Fe inclusion in the Wurtzite crystal structure of ZnO, a material with potential applications in spintronic devices.

1. Introduction

Transition metal (TM) doped oxide semiconductors at the nanoscale level have renewed interest in research due to the large number of properties that allowed expanding applications in different scientific and technological fields [1–3]. A particular case is the insertion of magnetic ions in the hexagonal structure of ZnO producing interesting electronic structures and relevant magnetic properties. Especially, pulsed Electron Paramagnetic Resonance (EPR) experiments probed that Fe^{3+} ions in ZnO nanostructures show ultra long spin coherence lifetimes, postulating the Fe:ZnO system as a highly promising spintronic system [4]. Fe^{3+} ions occupying the Zn site exhibit strong ferromagnetic (FM) interactions resulting from its high magnetic moment, controversially associated to the high, low or mixed spin configurations [5–7], $S = 5/2$ ($\mu_{eff} = 5.92\mu_B$) and $S = 1/2$ ($\mu_{eff} = 1.73\mu_B$), respectively. However, there are other factors involved in the observed magnetic interactions of

Fe-doped ZnO nanostructures, such as, nanoparticles (NPs) size, Oxygen and Zinc vacancies, etc [8].

One path that has been used to expand the functionalities and properties of ZnO nanostructures is by co-doping with magnetic and/or nonmagnetic ions [9–12]. Pascariu et al. [9] synthesized ZnO co-doped with Ni^{2+} and Co^{2+} by the co-precipitation method analyzing photocatalytic activities. Sharma et al. [13] reported the effect of co-doping with Co and Mn on the structural and optical properties of ZnO nanostructures and the magnetic properties in ZnO co-doped with Fe^{3+} and Co^{2+} ions [14]. Yakout et al. [15] also examined the ferromagnetism in ZnO nanocrystalline co-doped with Co and Mn. Yuan et al. [16] investigated the effect of pH on ZnO films co-doped with Fe and Co. According to the authors, magnetic properties of co-doped samples are related to oxygen vacancies and can be regulated through the sol pH. In the case of pure and Fe doped ZnO, the magnetic exchange mechanism involving intrinsic defects provoked by the pH variation lead to

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interesting magnetic properties [17]. Recently, Peña-Garcia et al. [18] analyzed the role of Y on the structural, magnetic and optical properties of Fe-doped ZnO NPs. They observed modifications in the hexagonal structure of ZnO caused by the inclusion of Fe and Y in the Zn sites. These elements provoke an increase of the intrinsic defects in the structure, generating ferromagnetic contributions combined with the paramagnetism of the compounds. Despite the various analysis presented to explain those results, there are questions about the origin of the induced defects due to the Y^{3+} inclusion that need to be clarified using other approaches.

Electron paramagnetic resonance (EPR) is a powerful technique to explore point defects in semiconductors, including bulk and nanostructured systems [8,19–22]. Specifically, the EPR spectrum of ZnO NPs evidences the presence of oxygen and zinc vacancies [23,24], being the main contributions for the observed EPR complex signal in this system, with a corresponding g value near to the free electron ($g_{\text{free}} = 2.0023$) [25–27]. Regarding to the stability of defects in ZnO NPs, it was reported that oxygen vacancies trapped by a single electron, V_O^+ , are associated to the core of the ZnO NPs because are stable contrasting with surface defects with a well defined g parameter ($g = 1.9588$) [26]. The Zn vacancies were also postulated as stable defects. Previous EPR studies revealed the presence of Fe^{3+} at the Zn (core), interstitial and surface sites of ZnO NPs [20].

For this study, the Electron Paramagnetic Resonance (EPR) technique was used as a fundamental tool to understand and determine intrinsic defects caused by the Y and Fe presence in the ZnO structure. The low concentration of Y (0.02) and Fe (0.00 $\leq y \leq 0.04$) used, prevented the formation and segregation of undesirable phases. Transmission Electron Microscopy (TEM) helped to determine the shape and size of the particles. Finally, the EPR spectra were simulated using the $S = 5/2$ spin Hamiltonian to corroborate the experimental results. This work contributes to understand the effects of ZnO nanostructures co-doping, which may have high repercussions for applications in spintronic devices [28].

2. Experimental details

2.1. Materials and synthesis of pure ZnO and co-doped with Y^{3+} and Fe^{3+}

For the synthesis of ZnO nanoparticles co-doped with Y/Fe, the sol gel method was used and specific details can be found in previous works [18,29,30]. Raw materials of analytical grade such as: $Zn(NO_3)_2$, Y (NO_3)₃, $Fe(NO_3)_3$ and citric acid, purchased from Sigma Aldrich Brazil (99.9%), were taken as initial precursors for the synthesis process. The general chemical formula employed for the samples was $Zn_{0.98-y}Y_{0.02}Fe_yO$ ($y = 0.00, 0.02, 0.03$ and 0.04) and were renamed as ZYF00 (undoped ZnO), ZYF20 (for $Zn_{0.98}Y_{0.02}O$), ZYF22 (for $Zn_{0.96}Y_{0.02}Fe_{0.02}O$), ZYF23 (for $Zn_{0.95}Y_{0.02}Fe_{0.03}O$) and ZYF24 (for $Zn_{0.94}Y_{0.02}Fe_{0.04}O$).

The soluble solutions of zinc nitrate (in distilled water), iron nitrate, yttrium nitrate were mixed with the citric acid solution, respecting the desired stoichiometric relationship. The mixture was magnetically stirred and heated at 95 °C for 5 h so as to obtain a viscous gel like solution. After this, a heat treatment at 130 °C (1 °C/min ramp) during 48 h was carried out and followed by a heating at 400 °C (1 °C/min ramp) for 1 h to burn-off the organic materials. Finally the samples were thermally treated at 700 °C (1 °C/min ramp) for 4 h to obtain the nanoparticles. A detailed analysis using X-ray diffraction, Fourier transform infrared (FTIR), Scanning Electron Microscopy, Vibrating Sample Magnetometer, Photoluminescence (PL) and Optical diffuse reflectance was reported by Peña-Garcia et al. in [18].

2.2. Physical methods

For this work, the morphology was studied using a transmission electron microscope (TEM), model JEOL JEM-1200 EX-II, with tungsten

filament electron gun, maximum electron energy of 120 keV and a Gatan Orius digital CCD capture system. Room-temperature EPR spectra were collected in a Bruker (Rheinstetten, Germany) Electron-Spin Resonance Spectrometer System instrument, EMX Micro 6"/1.2 kW, operating at X-band frequency. A standard Wilmad quartz tube in a high sensitive cavity was utilized. DPPH (α, α -diphenyl- β -picrylhydrazyl) was used for frequency calibration ($g = 2.0036$) at 300 K. Typical conditions in these measurements were: frequency 9.76 GHz, power 2 mW, 41.2×10^2 gain, 1 G modulation amplitude, time constant 2.56 ms, sweep time 30.72 s maintained for the five explored samples.

2.3. EPR simulations

The EPR spectra were simulated using the $S = 5/2$ spin Hamiltonian [31] ($3d^5$ for Fe^{3+}) given by Eq. (1), including axial, D , and rhombic, E , components to the zero-field splitting (ZFS) as follows:

$$\begin{aligned} \hat{H} = & g\mu_B \vec{H} \cdot \hat{S} + D \left(\hat{S}_z^2 - \frac{35}{12} \right) + E \left(\hat{S}_x^2 - \hat{S}_y^2 \right) \\ & + \frac{a}{6} \left(\hat{S}_x^4 + \hat{S}_y^4 + \hat{S}_z^4 - \frac{707}{16} \right) + \frac{7F}{36} \left(\hat{S}_z^4 - \frac{95}{14} \hat{S}_z^2 + \frac{81}{16} \right), \end{aligned} \quad (1)$$

where μ_B is the Bohr magneton, g is the Lande g -tensor and $\vec{H}$ is the external magnetic field. ξ, η, ζ denote the cubic electric field axes and z refers to the trigonal axis [31] at axial symmetry $E = 0$ which applies if the system has a C_n symmetry axis with $n \geq 3$. Single crystal parameters g, D, F and a were already reported for Fe^{3+} doped ZnO [31] and were used as a starting point in the present work. EPR spectra were simulated with the spin-Hamiltonian parameters using EasySpin [32].

3. Experimental results and discussion

Fig. 1 displays TEM images of the ZYF00 and ZYF20 samples (left up and down figures respectively) with their respective average particle size distributions, calculated from size counting on more than 100 NPs taken from several images. The average particle sizes distribution was estimated from the Gaussian fitting. The size dispersion is indicated in parentheses and was defined as the FWHM of the Gaussian distribution. A drastic reduction of the NPs size was observed for ZYF20 samples (9 (7) nm), compared to the ZYF00 sample (44(20) nm), demonstrating high sensitivity of the ZnO structure to the Y insertion, however, maintaining the structure in agreement with previous results in this system [18].

Fig. 2 shows TEM images for the ZYF22, ZYF23 and ZYF24 samples with their respective particle size distributions. As observed, the average particle sizes are: 9(7), 6(5) and 11(7) nm for the samples ZYF22, ZYF23 and ZYF24, respectively. These values are approximately 3 orders less than those obtained for the ZYF00 sample, indicating that the Fe^{3+} and Y^{3+} insertion in the ZnO crystal structure limits the growth of the crystallites due to the changes of the nucleation mechanisms during the ZnO crystallization in agreement with previous reports [33]. Other authors suggest that the effects of pH in Fe-doped ZnO samples contribute to the decrease in particle size [17]. The morphological study presented here are in correspondence with those reported by Peña-Garcia et al. [18] using X-ray diffraction.

Fig. 3 shows the measured EPR spectra at room temperature and $P_{\mu w} = 2$ mW for undoped, doped and co-doped ZnO NPs. The resonances were measured at the same experimental conditions (including the mass). All samples present a complex line-shape due to several intrinsic defects despite the diamagnetic nature of ZnO. To explore the complex role of Y^{3+} , it is appropriate to analyze and compare the ZYF00 and ZYF20 samples separately, before include the Fe^{3+} effects.

The EPR spectra of ZYF00 shows a very narrow resonance near to $g = 2$ and a broad line in the 2–3 kOe range. The broad line is related with uncoupled surface oxygen defects, producing a strong signal. This line is

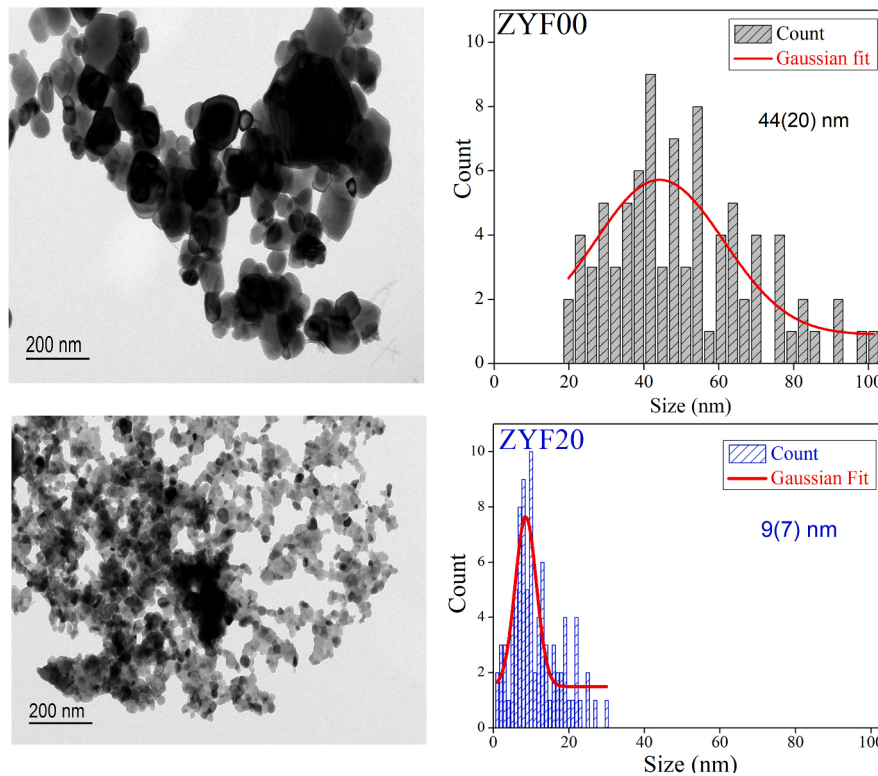


Fig. 1. TEM images of the ZYF00 and ZYF20 NPs using low magnification (left column). Particle size statistics from TEM images with histograms fitted to Gaussian distributions.

highly dependent on the annealing and size of the samples as evidenced in reference [8] and it will not be the focus of our study.

According to the literature, the narrow resonance is associated with oxygen vacancies (V_O) typically found in the hexagonal structure of ZnO [8]. Specifically, for ZYF00, the signal related to V_O^{+1} with $g = 1.946(1)$ (Figs. 3 and 4) is compatible with previous results for ZnO NPs annealed at 700 °C [8]. Comparing the obtained result for undoped ZnO (ZYF00) with the literature [8], strong differences in the measured line width, ΔH , were observed for V_O^{+1} ($\Delta H = 2.8$ Oe), reducing drastically the estimated ΔH in a factor of 0.2. This result may be associated with differences in the synthesis process and conditions. In addition, the drastic reduction can be associated to a lower variety of defects present in our samples, leading to a long relaxation time, compared with the annealed samples of ZnO reported by Çolak et al. [8]. Compatible ΔH values for V_O^{+1} were also observed in ZnO bulk samples ranging from 4.7 Oe at room temperature [21].

Fig. 3 also presents the line shape of the Y-doped ZnO sample (ZYF20), showing similarities with the ZYF00 sample. Nonetheless, for ZYF20, important differences were observed between 2000 Oe – 3000 Oe, which may be related to the size reduction of the NPs in agreement with Fig. 1 as well as, previous XRD studies [18]. This result is also consistent with the broadening of the EPR signal associated with surface effects [34]. At first glance, similar resonances are observed for the ZYF00 and ZYF20 samples close to $g = 2$ (dotted rectangle for ZYF00 and ZYF20 in Fig. 3).

Notwithstanding, for ZYF00 and ZYF20 samples, Fig. 4 shows the magnification of the region near to $g = 2$, specifically for the resonance associated with V_O^{+1} . It is possible to observe that the Y^{3+} inclusion in Zn^{2+} sites reduces the intensity and shift the resonance to higher magnetic fields. Using the Lorentzian fit of the line shape, it was possible to determine the g and ΔH parameters. For ZYF00 samples, it was obtained $g = 1.946(1)$, compatible with previous results [8] and $\Delta H = 2.8$ Oe, while for ZYF20 sample, we found $g = 1.942(1)$ and $\Delta H = 3.6$ Oe. This

small change can be understood as the modification of the antiferromagnetic local field due to intrinsic defects such as V_O^{+1} consistent with the diamagnetic ZnO matrix [18]. The enhancement of ΔH can be associated to a faster relaxation because the strong reduction of the NPs size by the Y^{3+} substitution (Fig. 1) compatible with observed broadening effects in the literature [34].

Keeping constant the Y concentration (0.02) and including Fe ions in the concentrations: 0.02, 0.03 and 0.04, it was possible to explore the configuration of 3d ions for co-doped ZnO NPs. The EPR signals for ZYF22, ZYF23, and ZYF24 samples are presented in Fig. 3. Describing the general behavior for the co-doped samples (Fig. 3), it is possible to observe modifications of the line shape when compared to ZYF00 and ZYF20, mainly for low fields (50 Oe – 2500 Oe). In addition, the resonance associated to V_O^{+1} observed for ZYF00 and ZYF20, disappears for co-doped samples. This result may be associated with the Fe^{3+} inclusion at the Zn^{2+} sites testifying the energetic competition between dopant elements within the hexagonal structure. Similar variations have also been reported by Peña-García et al. in [18] through optical and magnetic measurements.

Comparing the EPR spectra of ZYF22, ZYF23, ZYF24, no significant modifications in the general line shape were observed (Fig. 3). This result is in accordance with previous reports [19] for ZnO samples without Y^{3+} . From the literature, the EPR signal of Fe^{3+} for concentrations above 2.5% coalesces in an asymmetric broaden single resonance. This result is also compatible with an emergent ferromagnetic contribution observed by Peña-García et al. in [18] in magnetization measurements at room temperature for the ZYF22, ZYF22 and ZYF24 samples. An extra resonance with $g \approx 3.1$ appears for ZYF24 (denoted by * in Fig. 5) mainly associated to the size distribution of the measured NPs when compared with ZYF22 and ZYF23. This resonance will be better understood using EPR simulations.

Towards a proper physical description of the Y^{3+} substitution in the ZnO lattice coexisting with Fe^{3+} ions, it was necessary a comparative of

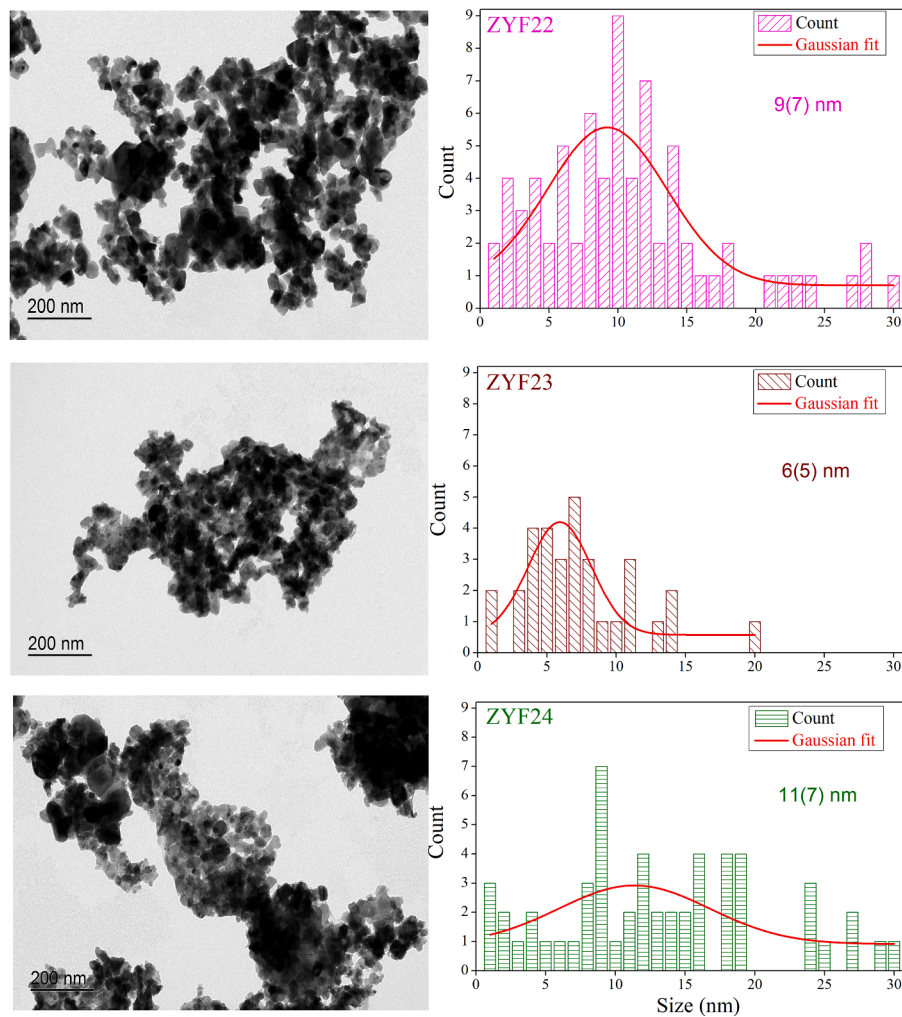


Fig. 2. TEM images of the ZYF22, ZYF23 and ZYF24 NPs using low magnification (left column). Particle size statistics from TEM images with histograms fitted to Gaussian distributions.

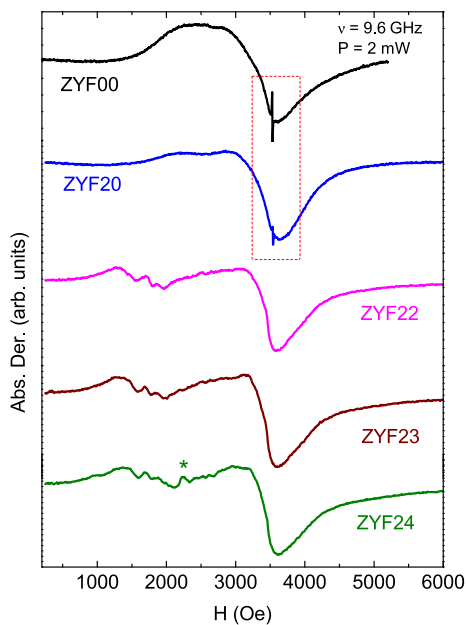


Fig. 3. EPR spectra of ZYF00 - ZYF24 at room-T and $P_{\mu w} = 2$ mW.

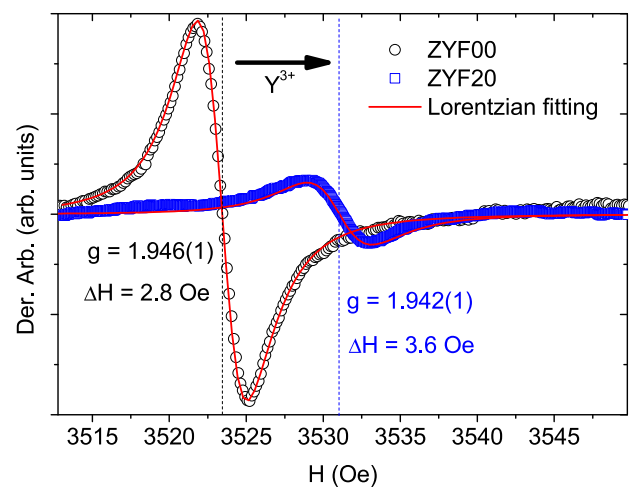


Fig. 4. EPR spectra of ZYF00 and ZYF20 with the extracted EPR parameters. Red line denotes the Lorentzian fitting of the line shape for both curves.

the line shape with a reference compound [20]. EPR experiments reported by Zhou et. al. [20] using ZnO NPs doped with 2% of Fe^{3+} were used to elucidate those effects, (see Fig. 5). A similar line shape was observed for all samples, which confirms the Fe^{3+} presence in the ZnO

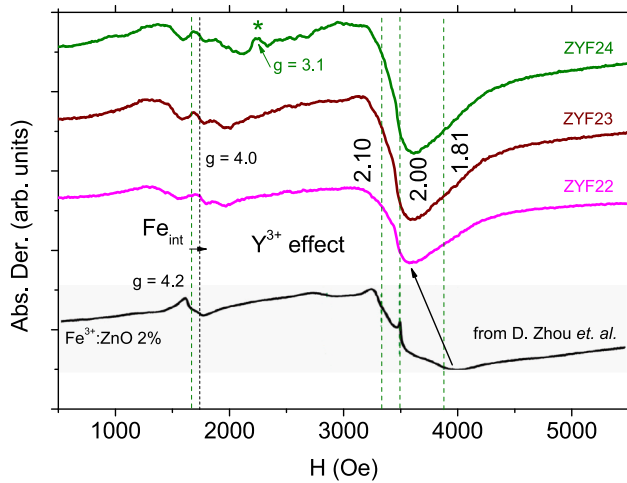


Fig. 5. EPR spectra of ZYF22, ZYF23 and ZYF24 compared with data of 2% doped Fe^{3+} ZnO NPs from D. Zhou [20] (black line).

structure.

Starting our comparative at low magnetic fields, the presence of a previously reported resonance near to $g = 4.2$ assigned to interstitial Fe^{3+} ions was also observed in our experiments (dotted lines) shifted to a higher magnetic fields ($g = 4.0$). This is caused by the Y^{3+} inclusion and independent of the Fe concentration as reported previously [19,20].

Comparing the central resonances near to $g = 2$, it is possible to observe an inhomogeneous broadening and the coalescence in a single resonance due to the Y^{3+} inclusion (Fig. 5 denoted as Y^{3+} effect). These effects are hard to be described because the many ingredients present in the co-doped NPs. Simulations of the line shape will elucidate the observed modifications of the Fe^{3+} line shape (narrowing/broadening) in our system. As a starting point, we used reported experimental parameters [31] of the Spin–Hamiltonian described in Eq. 1, to generate, the expected lines from Fe^{3+} in a single crystal of ZnO.

Fig. 6 (black line) shows simulated lines using the parameters: $D = -593.7 \times 10^{-4} \text{ cm}^{-1}$, $E = 0$, $a = 39 \times 10^{-4} \text{ cm}^{-1}$, $F = 4 \times 10^{-4} \text{ cm}^{-1}$, homogeneous line width (lw) = 30 Oe and isotropic $g = 2.0062$. In order to introduce known distortions in the simulated behavior [18,35],

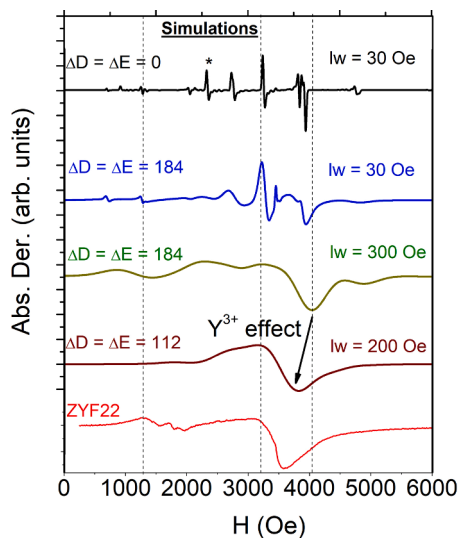


Fig. 6. Simulated EPR line shapes for $\Delta D = \Delta E = 0$ and $lw = 30$ Oe for the single crystal (black line), $\Delta D = \Delta E = 184 \times 10^{-4} \text{ cm}^{-1}$ and $lw = 30$ Oe (blue line), $\Delta D = \Delta E = 184 \times 10^{-4} \text{ cm}^{-1}$ and $lw = 300$ Oe (dark green) and $\Delta D = \Delta E = 112 \times 10^{-4} \text{ cm}^{-1}$ and $lw = 200$ Oe (wine line) compared with ZYF22 (red line).

expected for NPs, we use the following parameters: D -strain (ΔD), and E -strain (ΔE) still with 30 Oe of lw (blue line) following previous studies [20]. The simulated line shape (blue line) is compatible with the literature using 0.15% $\text{Fe}:\text{ZnO}$ colloidal nanocrystals [20]. However, strong differences appear when compared with the ZYF22 sample, corresponding to 2% of Fe^{3+} . This is expected, as a result of the absence of broadening effects in the simulation due to the exchange interactions in concentrated samples [19]. To include Fe-Fe interactions in the simulations it was introduced a homogeneous line broadening of 300 Oe, but maintaining the same strain parameters (green line). After this, the obtained line shape is compatible with previous reports in ZnO NPs doped with 2% of Fe^{3+} (see Fig. 5), but still different to the line shape of our representative sample ZYF22, since our sample contains an extra contribution from 2% of Y^{3+} . It is worth noting that the line width is related with the relaxation time and is strongly dependent to the magnetic interactions where a ferromagnetic response of the sample enhances their value. In the case of simulations, we assume a homogeneous value for simplicity, but it is expected inhomogeneities of this value in real systems. This condition would be behind the differences between simulations and experimental data. A considerable improvement of the simulated line shape was obtained when the strain and the linewidth decrease (wine line) to $112 \times 10^{-4} \text{ cm}^{-1}$ and 200 Oe respectively, simulating the Y^{3+} effect. With this, we believe that the Y^{3+} inclusion leads a local strain reduction and an impairment of the Fe-Fe interaction. This is expected in virtue of the inclusion of non magnetic ions in the core sites of NPs. It is worth noting that the replacement Y^{3+} (larger ions) by Zn^{2+} (smaller ions) supports the reduction of the Fe-Fe interaction in our simulations; in view of the local separation of the magnetic ions. The unknown resonance for ZYF24 (denoted by * in Fig. 5) can be related now to an expected line of Fe^{3+} in the core sites, denoting a large size distribution of the NPs and resulting in non-homogeneous Fe^{3+} line broadening, compatible with TEM results (see Fig. 2).

As a way to understand the aforementioned effects about to the Y^{3+} inclusion in the interstitial sites, Fig. 7 presents a simulated EPR line shape of Fe^{3+} with $E/D = 0.33$ and $E/D = 0.39$, corresponding to $g = 4.2$ and $g = 4.1$ respectively. Signals near to $g = 4.3$ are very often observed for Fe^{3+} in high-spin configuration [36]. This Fe^{3+} resonance was associated with rhombically distorted octahedral coordination environments at an interstitial site near the ZnO surface [20,36]. This correspond to $E/D \sim 3$ (a large rhombic component in the above described Spin–Hamiltonian). The observed shift in our experimental results shown in Fig. 5 now can be related to a possible modification of the rhombic component (denoting the transverse component of the magnetic dipole–dipole interaction). This result also confirms the presence of Y^{3+} in some interstitial sites. (see Fig. 7)

Finally, the ZYF22, ZYF23 and ZYF24 samples were reported with

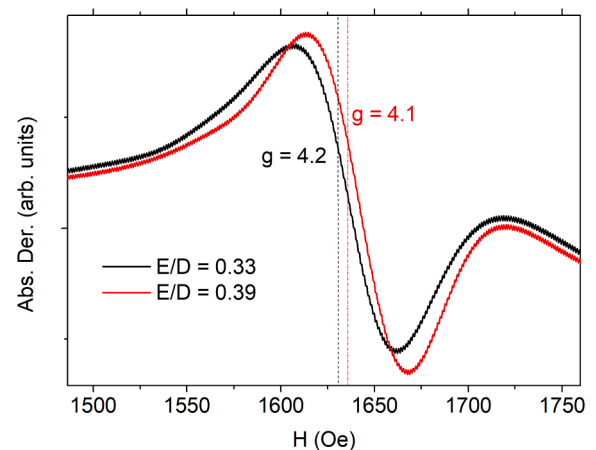


Fig. 7. Simulated EPR of Fe^{3+} with $E/D = 0.33$ and $E/D = 0.39$ corresponding with $g = 4.2$ and $g = 4.1$ respectively.

similar sizes, leading exchange interactions as the main contribution to the broadening of the resonances for concentrations above 2% of Fe^{3+} . With this, our simulations using large line widths are justified.

4. Conclusions

In summary, we present a comprehensive study using TEM and EPR on the effects of the Y^{3+} substitution in the $\text{Zn}_{0.98-y}\text{Y}_{0.02}\text{Fe}_y\text{O}$ ($y = 0.00, 0.02, 0.03$ and 0.04) system, using V_O^{+1} and Fe^{3+} as local probes. Transmission Electron Microscopy images showed agglomerated nanostructures, irregular shapes and average particle size distributions ranging from 6 nm to 44 nm in diameter. For pure ZnO sample, stable V_O^{+1} were observed in the EPR spectrum, while for the sample doped with Y a slightly modification in the resonance field and linewidth confirms the Y^{3+} presence in the Zn sites. From the EPR spectra of co-doped samples, we have also confirmed the existence of Fe^{3+} and Y^{3+} in the core Zn sites. The Y insertion reduces the lattice strain and the Fe-Fe interaction of the co-doped samples. Finally, the Y^{3+} was also confirmed in interstitial sites, shifting the resonance field to higher magnetic fields due to the enhancement of the rhombic component, E, of the Spin-Hamiltonian.

CRedit authorship contribution statement

M. Cabrera-Baez: Conceptualization, Writing - original draft, Data curation, Formal analysis. **E. Padrón-Hernández:** Funding acquisition, Conceptualization, Writing - original draft, Data curation, Formal analysis. **João M. Soares:** Data curation, Formal analysis. **F.E. P. Santos:** Data curation, Formal analysis. **Y. Guerra:** Conceptualization, Formal analysis. **R. Peña-García:** Funding acquisition, Conceptualization, Writing - original draft, Data curation, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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